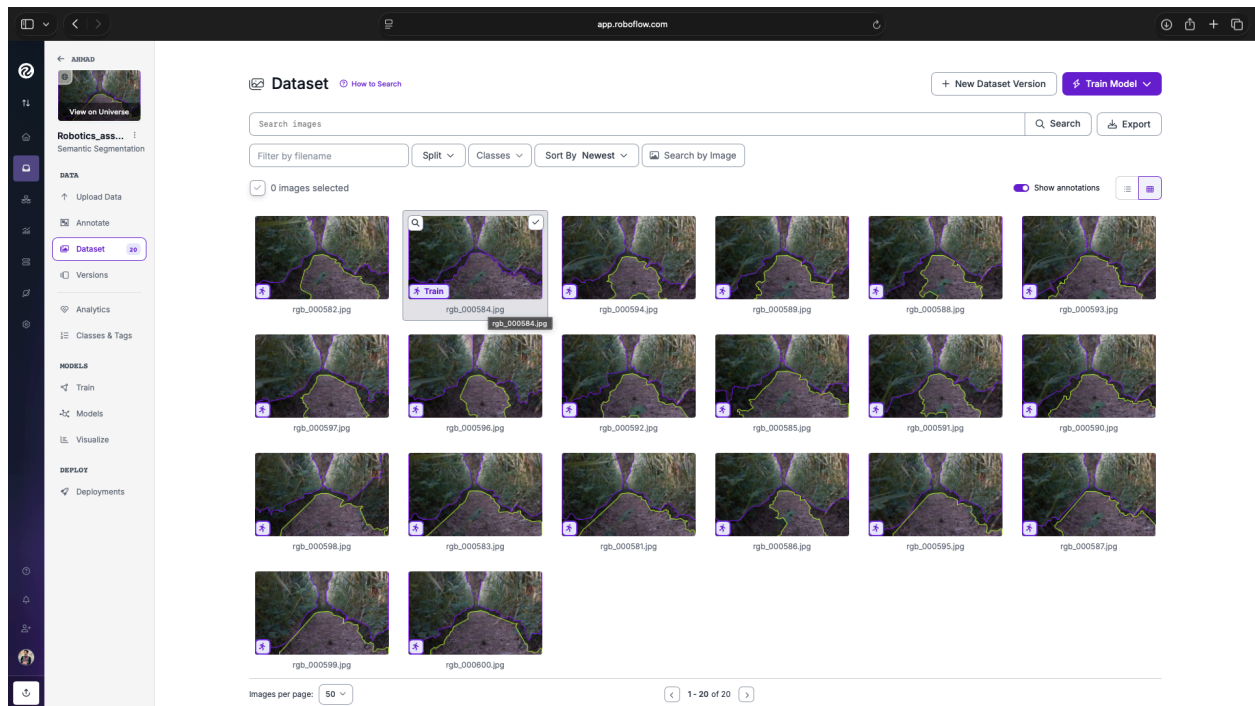


Muhammad Ahmad
25280065

TASK 1

1

First, the dataset images were annotated using the **Roboflow editor**. Polygon annotations were used to label the objects in the images. While annotating, special care was taken to ensure that the polygons closely follow the boundaries of the objects so that the segmentation masks would be accurate.



2

After applying the augmentation settings in Roboflow, the dataset generated a total of **60 images**. The screenshot below shows the dataset with the final image count after augmentation.

_classes.csv

rgb_000581_j...ed_mask.png
rgb_000581_j...fa773f7ed.jpg
rgb_000581_j...6f_mask.png
rgb_000581_j...fec17d96f.jpg
rgb_000581_j...5e_mask.png
rgb_000581_j...e43e3c5e.jpg
rgb_000582_...f97_mask.png
rgb_000582_...495e3f97.jpg
rgb_000582_...6e_mask.png
rgb_000582_...a629926e.jpg
rgb_000582_...ab_mask.png
rgb_000582_...baf2b0ab.jpg
rgb_000583_...26_mask.png
rgb_000583_...a522e926.jpg
rgb_000583_...52_mask.png
rgb_000583_...3b6a5352.jpg
rgb_000583_...85_mask.png
rgb_000583_...70987b85.jpg
rgb_000584_...40f_mask.png
rgb_000584_...6a7de140f.jpg
rgb_000584_...ab_mask.png
rgb_000584_...1cbb5eab.jpg
rgb_000584_...d1_mask.png
rgb_000584_...213caead1.jpg
rgb_000585_...c9_mask.png
rgb_000585_...f832212c9.jpg
rgb_000585_...43_mask.png
rgb_000585_...93101543.jpg
rgb_000585_...68_mask.png
rgb_000585_...342dad68.jpg
rgb_000586_...35_mask.png
rgb_000586_...71e2c8135.jpg
rgb_000586_...6f7_mask.png
rgb_000586_...254db6f7.jpg
rgb_000586_...61_mask.png
rgb_000586_...66ff18261.jpg
rgb_000587_j...d6_mask.png
rgb_000587_j...1fd1758d6.jpg
rgb_000587_j...96_mask.png



120 items

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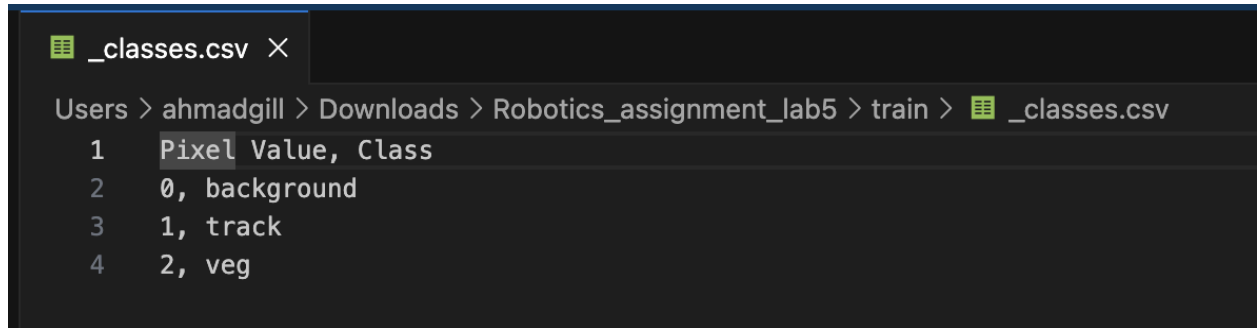
Create PDF



More...

3

Class. Csv is this



```
_classes.csv X
Users > ahmadgill > Downloads > Robotics_assignment_lab5 > train > _classes.csv
1 Pixel Value, Class
2 0, background
3 1, track
4 2, veg
```

Task 4: Submit the Python code used to convert the raw integer arrays (0, 1, 2) into the RGB color palette (Red, Green, Black).

```
import os
import numpy as np
import cv2

folder="/Users/ahmadgill/Documents/GitHub/LUMS-Academics/Second
Semester/RoboTics641/Lab5/Robotics_assignment_lab5/train"
output=os.path.join(folder,"visualized")
os.makedirs(output,exist_ok=True)

palette={0:[0,0,0],1:[0,255,0],2:[255,0,0]}

for f in os.listdir(folder):
    if "_mask.png" not in f:
        continue

    mask_path=os.path.join(folder,f)
    image_name=f.replace("_mask.png",".jpg")

    mask=cv2.imread(mask_path,cv2.IMREAD_GRAYSCALE)
```

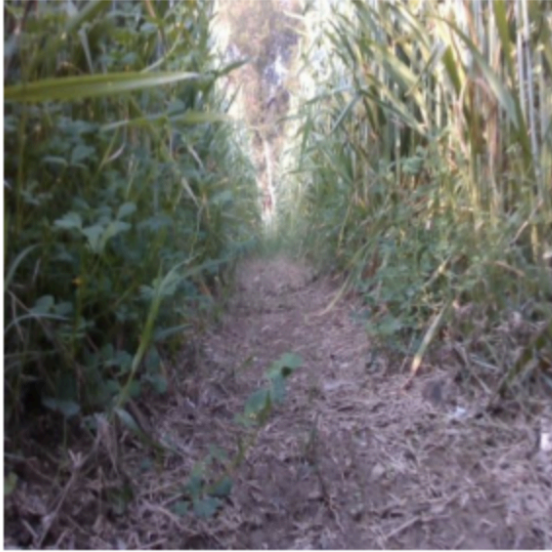
```
color=np.zeros((mask.shape[0],mask.shape[1],3),dtype=np.uint8)

    for k,v in palette.items():
        color[mask==k]=v

cv2.imwrite(os.path.join(output,image_name),cv2.cvtColor(color,cv2.COLOR_RGB2BGR))
```

Task 4: Provide a screenshot showing an original RGB image side-by-side with your newly colored mask.

Original RGB Image



Colorized Mask



Original RGB Image



Colorized Mask

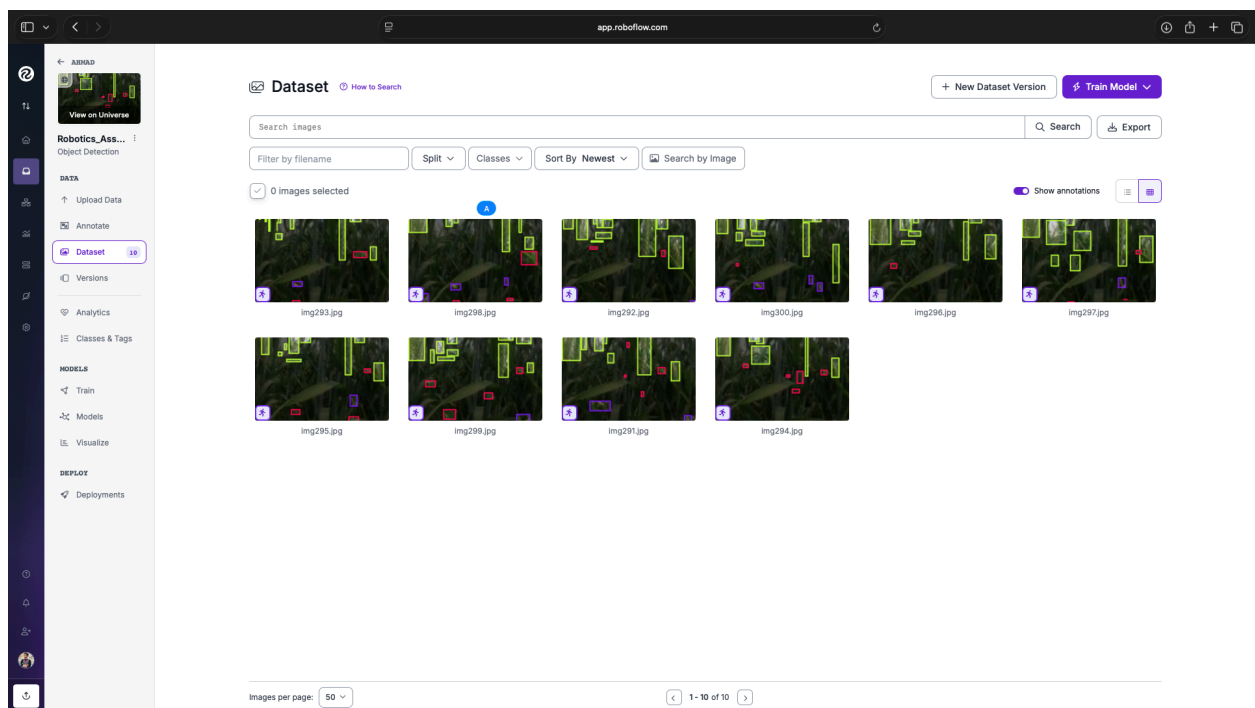


TASK2

Task 1: Dense Annotation in Roboflow

For Task 1, the dataset images were annotated using the **Roboflow annotation editor**. Each image was labeled carefully using polygon annotations. While performing the labeling, attention was given to ensure that the polygons closely follow the boundaries of the objects. This helps create accurate segmentation masks and improves the quality of the dataset.

The annotations were also made **densely**, meaning that as many relevant objects as possible were labeled in each image. All visible instances belonging to the required classes were annotated to ensure that the dataset is complete and suitable for training.



Task 2: YOLO Annotation Format

The bounding box annotations in this dataset follow the **YOLO 5-column format**. Each line in the label file represents one object in the image and contains five values:

- **class_id** represents the category of the object.
- **x_center** represents the x-coordinate of the center of the bounding box.
- **y_center** represents the y-coordinate of the center of the bounding box.
- **width** represents the width of the bounding box.
- **height** represents the height of the bounding box.

All coordinate values are **normalized between 0.0 and 1.0**. This means the values are expressed as a fraction of the image width and height rather than absolute pixel values.

Normalization is important because it allows the annotations to remain consistent across images of different sizes. Since the values are relative to the image dimensions, the same annotation format can be used regardless of the image resolution.

Task 3: Ground Truth Bounding Box Visualization

To verify that the bounding box annotations were exported correctly, a Python script was written to visualize the ground truth boxes on the images. The script reads each image along with its corresponding .txt label file and converts the normalized YOLO coordinates into pixel coordinates.

```
import os
import cv2
import matplotlib.pyplot as plt

base="/Users/ahmadgill/Documents/GitHub/LUMS-Academics/Second
Semester/RoboTics641/Lab5/Robotics_AssignmentLab5Final/train"

image_folder=os.path.join(base,"images")
label_folder=os.path.join(base,"labels")

colors={
2:(255,0,0),
1:(0,255,0),
0:(0,0,255)
}

names={
0:"diseased",
1:"pests",
2:"wheat"
```

```
}

for f in os.listdir(image_folder):

    if not f.endswith(".jpg"):
        continue

    img_path=os.path.join(image_folder,f)
    txt_path=os.path.join(label_folder,f.replace(".jpg",".txt"))

    if not os.path.exists(txt_path):
        continue

    img=cv2.imread(img_path)
    h,w,_=img.shape

    with open(txt_path) as file:
        lines=file.readlines()

    for line in lines:

        c,xc,yc,bw,bh=map(float,line.split())
        c=int(c)

        xmin=int((xc-bw/2)*w)
        ymin=int((yc-bh/2)*h)
        xmax=int((xc+bw/2)*w)
        ymax=int((yc+bh/2)*h)

        cv2.rectangle(img, (xmin,ymin), (xmax,ymax), colors[c],2)
```

```
cv2.putText(img,names[c],(xmin,ymin-5),cv2.FONT_HERSHEY_SIMPLEX,  
0.5,colors[c],1)
```

```
img=cv2.cvtColor(img,cv2.COLOR_BGR2RGB)
```

```
plt.figure(figsize=(6,6))
```

```
plt.imshow(img)
```

```
plt.axis("off")
```

```
plt.title("Ground Truth Bounding Boxes")
```

```
plt.show()
```

```
break
```

Ground Truth Bounding Boxes

